Comp683: Computational Biology

Lecture 13

March 9, 2026

Today

- Contrastive PCA for dealing with background data.
- Integrating multiple single-cell datasets via conos

Announcements

- Please sign up with your project group here by today `https://docs.google.com/document/d/13c8XkcrTe-nCmwiF3utoyvZQfPmigCGPsoYh0JHGhtg/edit?usp=sharing`. Proposal template available in git repo.

Problem : removing a background signal in a biological dataset

- Consider high-dimensional gene expression measurements collected from people from all over the world.
- Suppose these patient samples also correspond to healthy and cancer patients.
- If the question is to find gene expression patterns associated with cancer subtypes, PCA on our samples may mostly reflect demographic variation between patients, rather than biological variation related to cancer subtypes.

Intro to Contrastive PCA

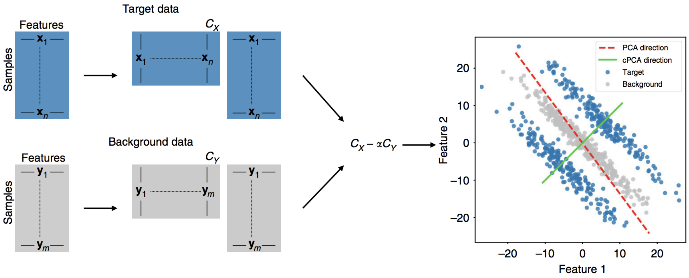

Figure: from Abid *et al.* Nature Communications. 2018. When projecting the data, the goal is to find the target direction that has the highest variance in the target data in comparison to the background data.

Thinking About Background Data

- Given two groups of datapoints (e.g. patient measurements), you can imagine there is variance common to both datasets and variance characteristic of each one.
- For example, thinking about a control group and a disease group, both have population-level variation, but the disease group has particular disease subtypes.
- As another example, consider time series data when you want to decouple variation from a particular timepoint from variation across the entire time series.
- Choice of background dataset is important here and should ideally contain ‘structure‘ that we would like to remove form the target data.

Synthetic MNIST Example

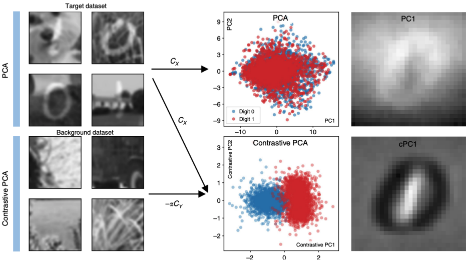

Figure: MNIST 1s and 0s are overlaid on scenes. A background dataset of grass is introduced, which allows better separation of 1s and 0s. In **c**, bright pixels shown the contribution to PCs or cPCs.

Motivating Biological Examples

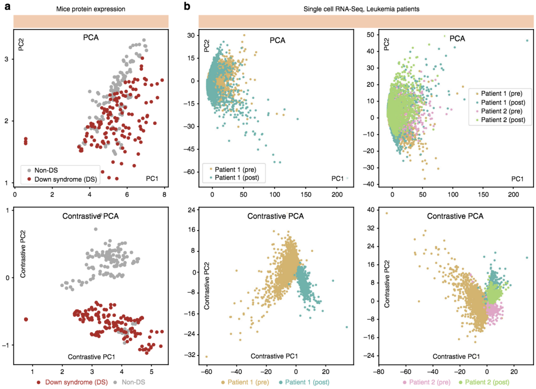

Figure: from Abid *et al.* Nature Communications. 2018. (Left) : Protein expression in Down Syndrome vs Non Down Syndrome Mice. Background data are control mice that have not been exposed to shock therapy. (Right) scRNA-seq of pre and post transplant, with healthy donor as background.

cPCA Problem Setup

- Assuming we start with d-dimensional target data $\{\mathbf{x}_i \in \mathbb{R}^d\}$ background data $\{\mathbf{y}_i \in \mathbb{R}^d\}$

For some direction vector, $\mathbf{v} \in \mathbb{R}^d_{\text{unit}}$ the variance it accounts for in the target and background data can be expressed as,

$$\text{Target data variance : } \lambda_X(\mathbf{v}) \stackrel{\text{def}}{=} \mathbf{v}^T \mathbf{C}_X \mathbf{v}$$

$$\text{Background data variance : } \quad \lambda_Y(\mathbf{v}) \stackrel{\text{def}}{=} \mathbf{v}^T \mathbf{C}_Y \mathbf{v}$$

What is happening here and what does this remind you of?

Given a contrast parameter $\alpha \geq 0$ that quantifies the trade-off between having high target variance and low background variance, cPCA computes the contrastive direction $\mathbf{v}$ by optimizing

$$\mathbf{v}^* = \text{argmax}_{\mathbf{v} \in \mathbb{R}^d_{\text{unit}}} \ \lambda_X(\mathbf{v}) - \alpha \lambda_Y(\mathbf{v})$$

This problem can be rewritten as

$$\mathbf{v}^* = \text{argmax}_{\mathbf{v} \in \mathbb{R}^d_{unit}} \mathbf{v}^T \left(C_X - \alpha C_Y \right) \mathbf{v}$$

cPCA is Quite Simple!

Algorithm 1 cPCA for a Given α

Inputs: target data $\{\mathbf{x}_i\}_{i=1}^n$; background data $\{\mathbf{y}_i\}_{i=1}^m$; contrast parameter α; the number of components k.

Centering the data $\{\mathbf{x}_i\}_{i=1}^n$, $\{\mathbf{y}_i\}_{i=1}^m$.
Calculate the empirical covariance matrices:

$$C_X = \frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_i \mathbf{x}_i^T, C_Y = \frac{1}{m}\sum_{i=1}^{m} \mathbf{y}_i \mathbf{y}_i^T.$$

Perform eigenvalue decomposition on

$$C = (C_X - \alpha C_Y).$$

Compute the the subspace $V \in \mathbb{R}^k$ spanned by the top k eigenvectors of C.
Return: the subspace V.

Figure: Just do eigendecomposition on **C** and consider the eigenvectors corresponding to the top k eigenvalues of C.

For $\alpha = 0$, cPCA will create directions that maximize the target variance. For higher α, directions with smaller background variance become more important.

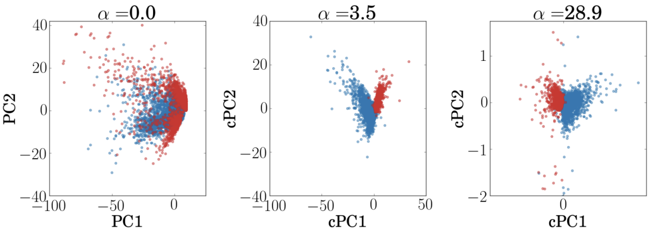

Figure: from Abid *et al.* Nature Communications. 2018. This dataset is visualizing cells from two different samples.

Recap

We have now seen Meld vs Cydar vs Milo vs cPCA (if we have a background). Thoughts? What would you do to compare your samples from disparate clinical or experimental groups?

Combining Multiple Single-Cell Datasets

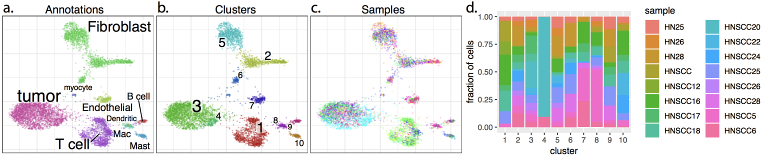

Figure: from Barkas *et al.* Nature Methods. 2019. Conos looks at how to integrate cells from multiple datasets (patients, tissues, etc.)

- The problem is a bit different from batch effect effect correction where you can identify technical artifacts and get rid of them. Cell-populations might be completely missing from particular datasets.

Conos Overview: Construct a Joint Between-Cell Graph

The goal is to establish a unified graph representation of the multiple single-cell datasets. Specifically, to infer cell-populations across all datasets, Conos seeks to infer inter-cell edges between datasets.

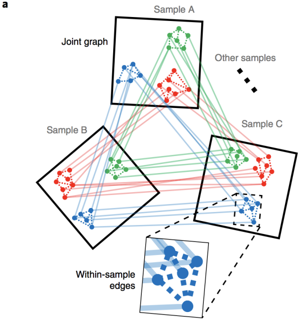

Figure: from Barkas *et al.* Nature Methods. 2019.

Pairwise Dataset Alignment

- As a pre-processing step, choose a set of high-variance genes. (The authors use 2,000).
- For a pair of datasets, i and j, let G_i and G_j denote their corresponding set of features measured per cell. Then consider only features that are measured in both datasets (so $G_i \cap G_j$)
- The similarity between cells K and l in datasets i and j is $w_{kl} = \exp(-\frac{||M_k^i - M_l^j||}{\sigma})$

Creating the Joint Graph

- Use w_{kl} for k-NN gaphs
- For **inter-sample edges**, connect each cell to its 15 nearest neighbors by default
- For **intra-sample edges**, connect each cell to its 5 nearest neighbors.
- Create joint clusters by clustering the graph with a graph-partitioning method.

Controlling Mixing Between Datasets

- Add a k_1 parameter or mixing parameter that allows for an increase of the nearest neighbor search radii, k. Control k_1 with an alignment strength parameter, $k_1 = \alpha^2 K_{\mathsf{max}}$ (K_{max} is the maximum number of total cells across samples.

α ranges between 0 and 1 and 0 corresponds to alignment with no additional edges, and 1 corresponds to a full alignment.

This is followed by a pruning step...

They have a little strategy to reduce maximal degree closer to k and to make the graph less dense.

- Order nodes from highest to lowest degree
- For each node, order edges by the degree of target vertices (high to low)
- Algorithm goes through nodes and corresponding edges and removes an edge if the degrees of both incident nodes are larger than a specific cutoff, k_0

Rebalance Edge Weights

- Since samples are often collected across conditions, the authors wanted to provide flexibility to control how likely pairs of cell populations are to be mapped to each other, between conditions. Specifically, balance edge weights between cells connected between the same or different values of a factor (e.g. experimental vs control or experiment type.).

The solution is to minimize the following,

$$\sum_{l=1}^{N_{\text{factors}}} \sum_{s=1}^{N_{\text{cells}}} \left| \frac{\sum_{t \in \text{adj}_l(s)} w_{st}}{\sum_{t \in \text{adj}(s)} w_{st}} - \frac{1}{N^s_{\text{factors}}} \right|$$

Unpacking...

$$\sum_{l=1}^{N_{\text{factors}}} \sum_{s=1}^{N_{\text{cells}}} \left| \frac{\sum_{t \in \text{adj}_l(s)} w_{st}}{\sum_{t \in \text{adj}(s)} w_{st}} - \frac{1}{N^s_{\text{factors}}} \right|$$

- N_{factors} is the total number of factor levels
- N_{cells} is the total number of cells.
- $adj(s)$ is the set of cells adjacent to cell s.
- $adj_l(s)$ is the set of cells adjacent to s and belong to factor level l.
- w_{st} is the weight of the edge between cells s and t
- N^s_{factors} is the number of different factors of cells connected to s.

Imbalance Between Factor l and cell s

For their minimization they first estimate the imbalance ratio for a cell s and a factor level, l as,

$$u_{sl} = N^{s}_{\text{factors}} \frac{\sum_{t \in \text{adj}_l(s)} w_{st}}{\sum_{t \in \text{adj}(s)} w_{st}}$$

Using Imbalance to Update Edge Weights

Edge weights are updated using the imbalance computed in the previous slide as,

$$w_{st} = \frac{w_{st}}{\sqrt{u_{sl_t} u_{tl_s}}}.$$

- Here l_c denotes the factor level of cell c.
- This process is repeated 50 times.

Effect of Alignment Strength

Here is an example varying alignment strength on a dataset containing cells from multiple technologies.

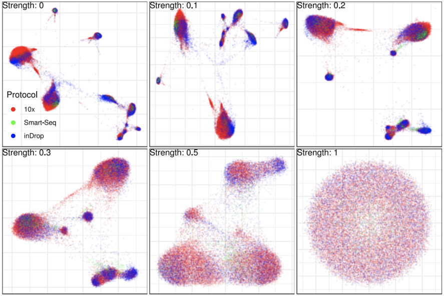

Figure: from Barkas *et al.* Nature Methods. 2019.

Example 1: Bone Marrow and Chord Blood

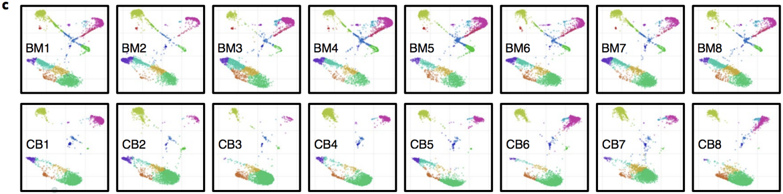

Figure: from Barkas *et al.* Nature Methods. 2019. You can see similarities and differences between cell-populations in each dataset.

Visualizing the Joint Graph

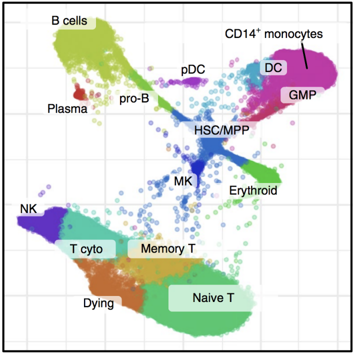

Figure: from Barkas *et al.* Nature Methods. 2019. The layout of the joint graph is determined by LargeVis.

Predicting a Cell's Label from the Graph Structure

Label propagation can be used to predict a cell's label based on the labels of neighboring cells.

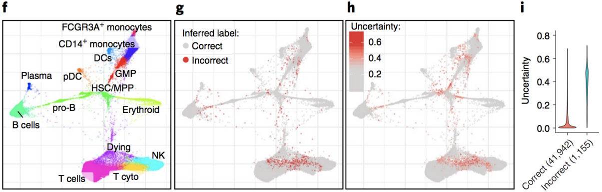

Figure: from Barkas *et al.* Nature Methods. 2019. Cells colored red represent those with incorrect predictions.

Switching Gears $\rightarrow$ Issues We Have Seen

Consider a mass cytometry or scRNA seq dataset from a huge clinical cohort (hundreds of samples, each with 100s of thousands of cells)

- We can't deal with all of these cells at once (clustering, visualization, etc)
- Downsampling is biased towards the higher frequency cell-types
- **Compression:** What if we want to take a representative subsample of the entire dataset?

An Important Task from a Practical Viewpoint: Patient Outcome Prediction

If you want to build a regression model, get a prediction for each cell and pool predictions for cells across patients, you are limited by the number of cells

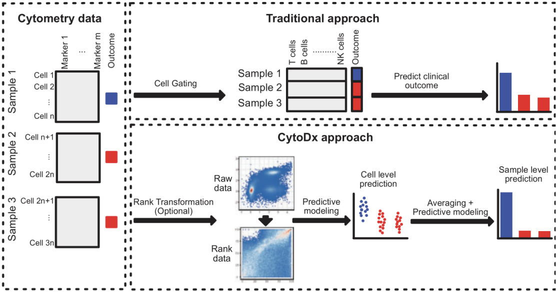

Figure: from Hu *et al.* Bioinformatics. 2018.